機械力学 I 解答用紙

week - 10 非減衰強制振動，減衰強制振動

学籍番号 20T1126N　氏名 YANG GUANGZE

m11a03C0201

(1)	$m\ddot{x} + kx = F\cos\omega t$
(2)	$x = \frac{F}{-m\omega^2+k} \cdot \cos\omega t$

m11b02C0102

(1)	$X = \frac{F}{\sqrt{(-m\omega^2+k)^2+(c\omega)^2}} \approx 0.306$ [mm]
(2)	$\phi = \mathrm{atan2}(c\omega, -m\omega^2+k) \approx 134$ [deg]

フリースピース（質問とか）

m11b04C0301

(1)	$\zeta = \frac{c}{2\sqrt{mk}} \approx 0.012$
(2)	$\omega_n = \sqrt{\frac{k}{m}} \approx 63.2$ [rad/s]
(3)	$r = \frac{\omega}{\omega_n} \approx 2.00$
(4)	$M_d = \frac{1}{\sqrt{(1-r^2)^2+(2\zeta r)^2}} \approx 0.333$
(5)	$\phi = \mathrm{atan2}(2\zeta r, 1-r^2) = 127$ [deg]

m11b04C0401

(1)	$M_{dp} = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \approx 5.03$
(2)	$X_{st} = \frac{F}{k} = 5.00 \times 10^{-5}$ [m]
(3)	$X = M_{dp} \cdot X_{st} = 2.52 \times 10^{-4}$ [m]
(4)	$\omega_n = \sqrt{\frac{k}{m}} = 200$ [rad/s]
(5)	$\omega = \sqrt{1-2\zeta^2}\,\omega_n \approx 198$ [rad/s]